

Technical Design Document (TDD)

Logistics ERP Analytics

1. Architecture Overview

Solution uses a layered architecture: Data Source (Excel/ERP) → Data Processing (SQL/Python) → Data Model (Power BI) → Visualization (Dashboards).

2. Data Sources

Primary source: logistics_erp Excel dataset. Contains customer, product, order, shipment, and inventory data.

3. Data Ingestion

Data is imported into Power BI using Power Query. Data cleaning includes removing nulls, formatting dates, and standardizing columns.

4. Data Modeling

Star schema design:

- Fact Tables: Orders, Shipments, Inventory
- Dimension Tables: Customer, Product, Supplier, Warehouse, Date

Relationships created using primary and foreign keys.

5. ETL Process

Extract: Load data from Excel

Transform: Clean, normalize, create calculated columns

Load: Store in Power BI model

6. DAX Calculations

- Total Orders = COUNT(Order ID)
- Total Revenue = SUM(Order Value)
- Avg Delivery Time = AVERAGE(DATEDIFF(Dispatch Date, Delivered Date))
- Inventory Turnover = SUM(Issued Qty) / AVERAGE(Closing Qty)

7. Dashboard Design

Dashboards created for CEO, CFO, COO, Warehouse, Procurement, Logistics, Inventory, Risk, Sales.

Includes charts, KPIs, slicers, and drill-through pages.

8. Security

Row Level Security (RLS) implemented to restrict access by role (e.g., warehouse-specific access).

9. Performance Optimization

- Use aggregated tables
- Enable incremental refresh
- Optimize DAX measures
- Use indexing in SQL if applicable

10. Deployment

Publish Power BI report to Power BI Service.

Set up scheduled refresh and workspace access.

11. Monitoring

Track refresh failures, performance metrics, and user activity using Power BI Service.

12. Backup & Recovery

Maintain backup of dataset and reports.

Use version control for report files (.pbix).